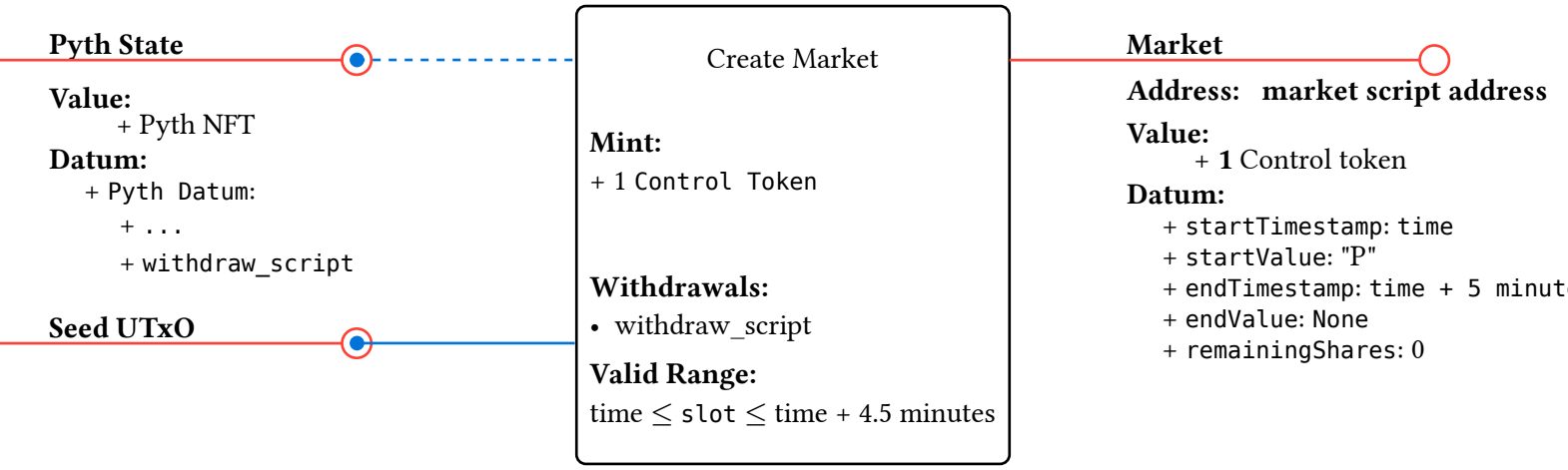
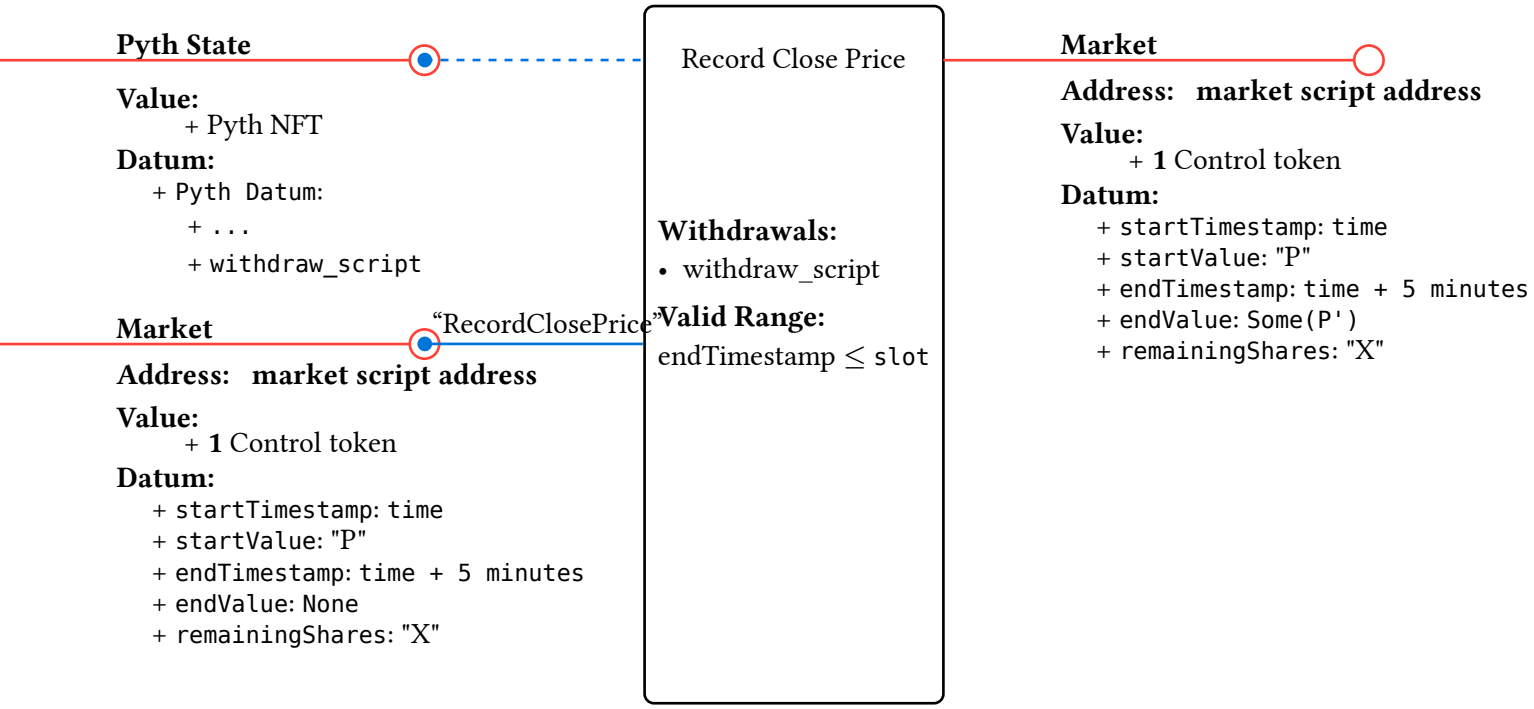
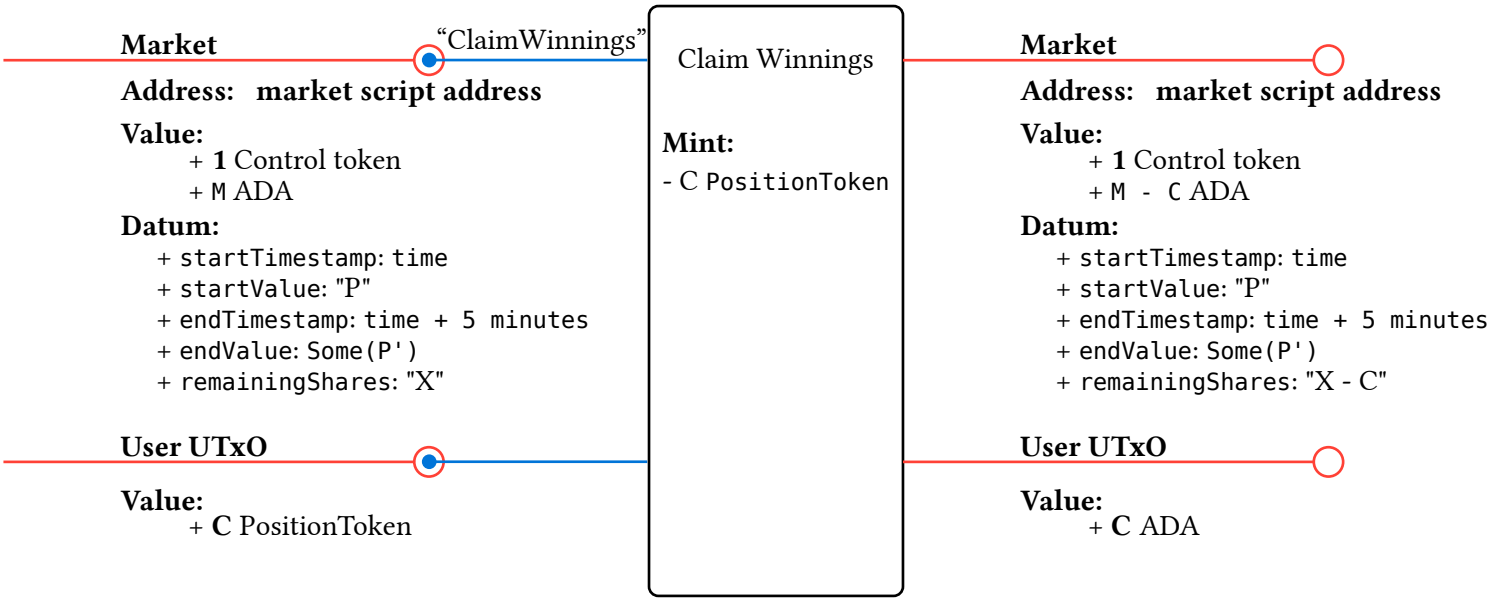


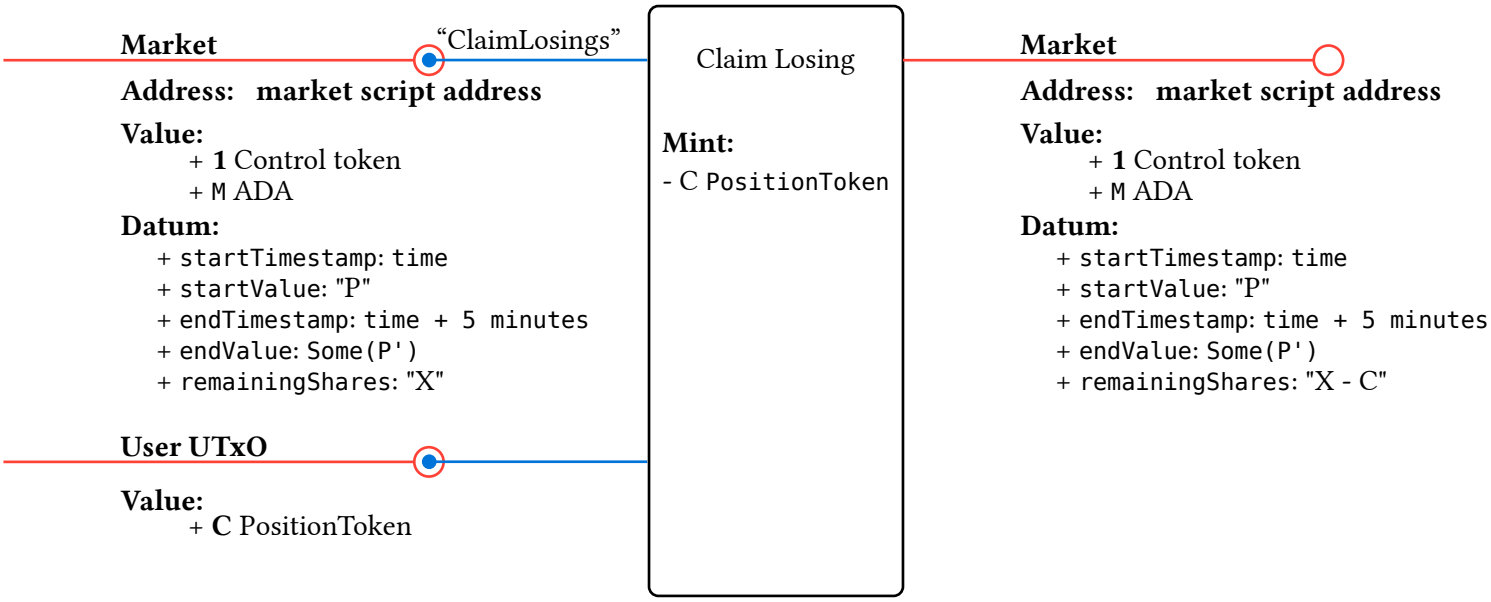
Notes: startValue should be provided by the oracle



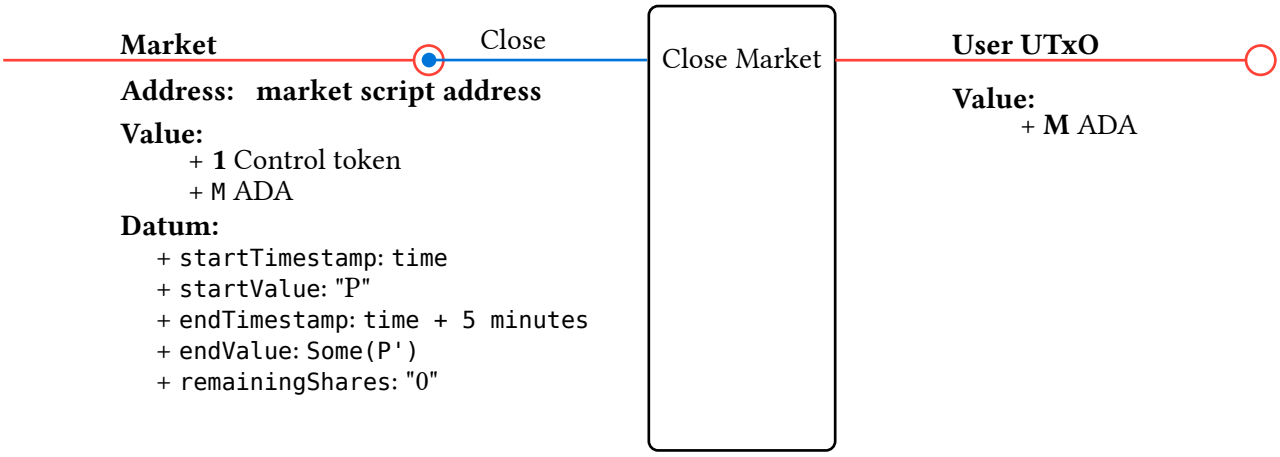
Notes: endValue should be provided by the oracle

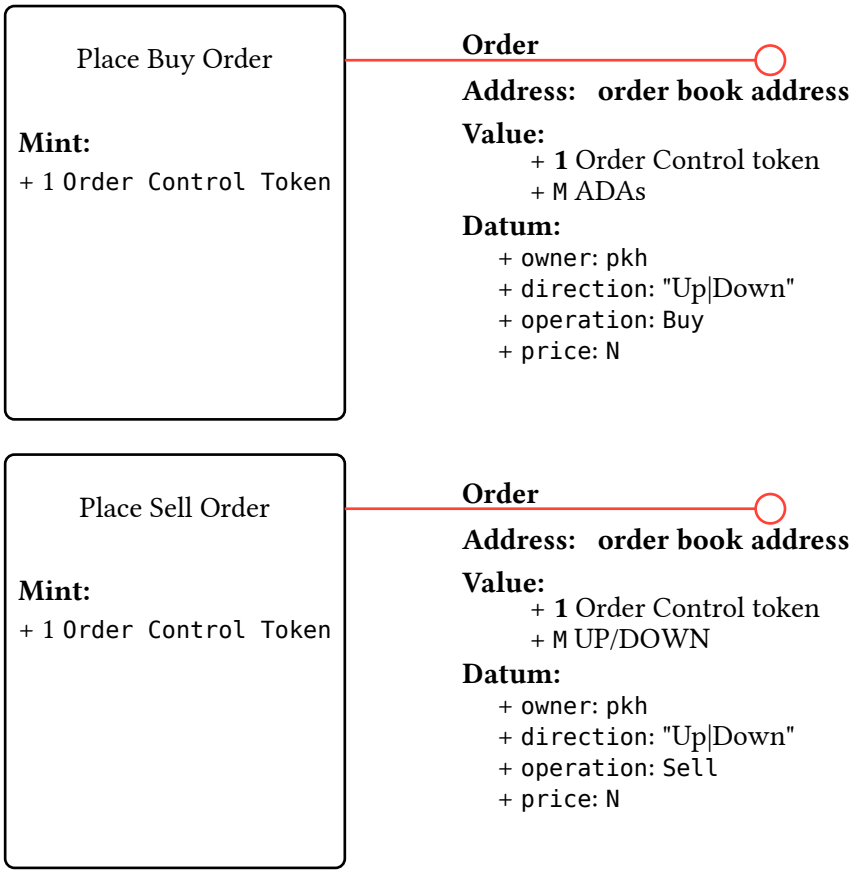


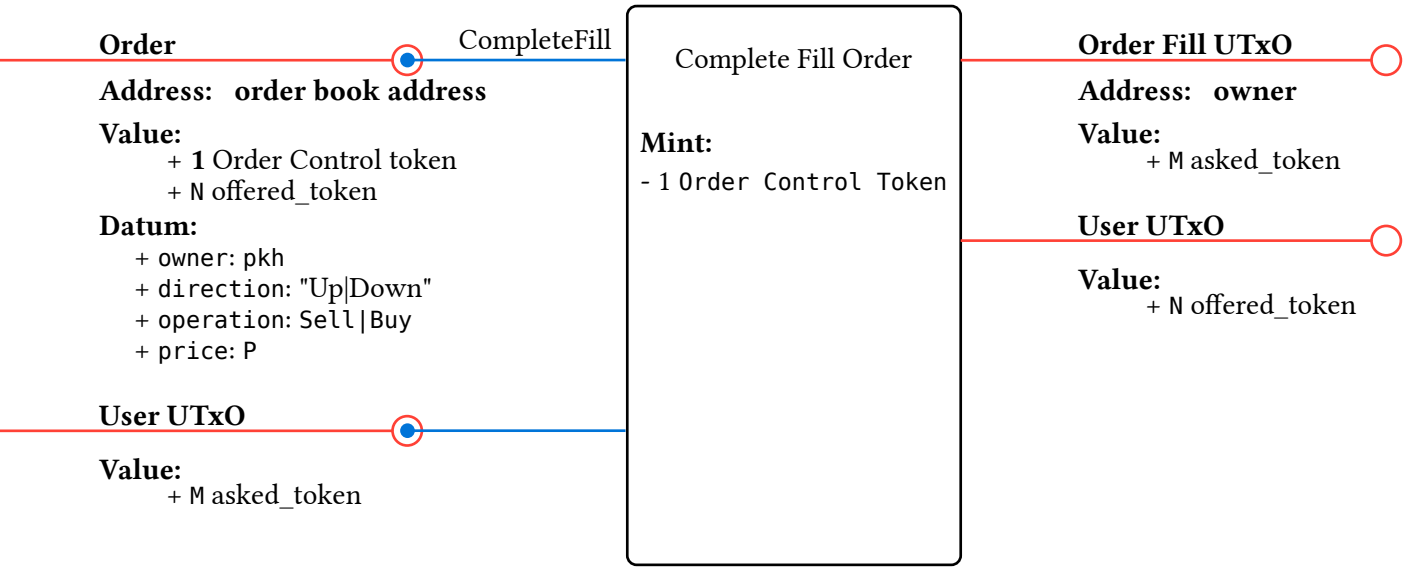
Notes: if P' >= P { PositionToken = "UP" }
else { PositionToken = "DOWN"}



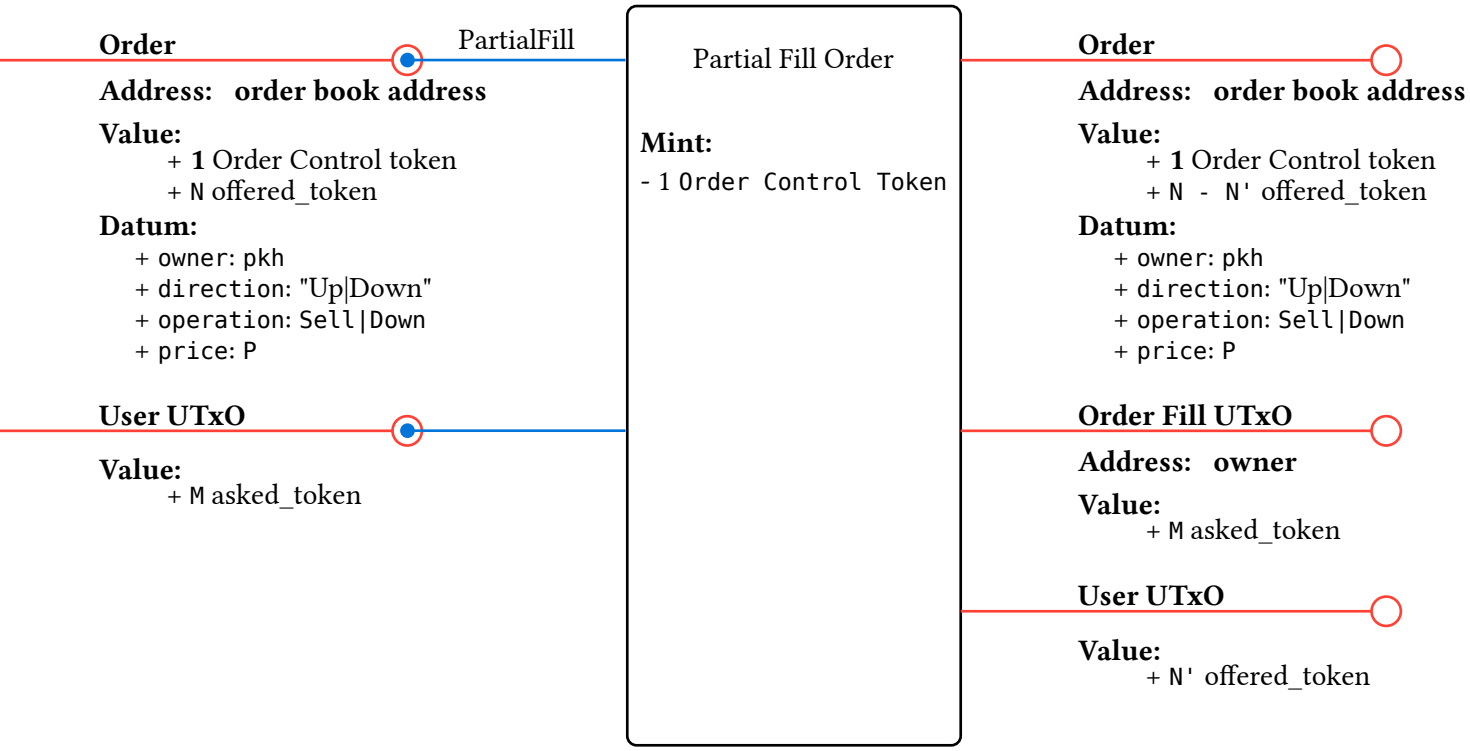
Notes: if P' < P { PositionToken = "UP" }
else { PositionToken = "DOWN"}


Market
Address: market script address
Value:
+ 1 Control token
+ M ADA
Datum:
+ startTimestamp: time
+ startValue: "P"
+ endTimestamp: time + 5 minutes
+ endValue: Some(P')
+ remainingShares: "0"
Close
Close Market
User UTxO
Value:
+ M ADA

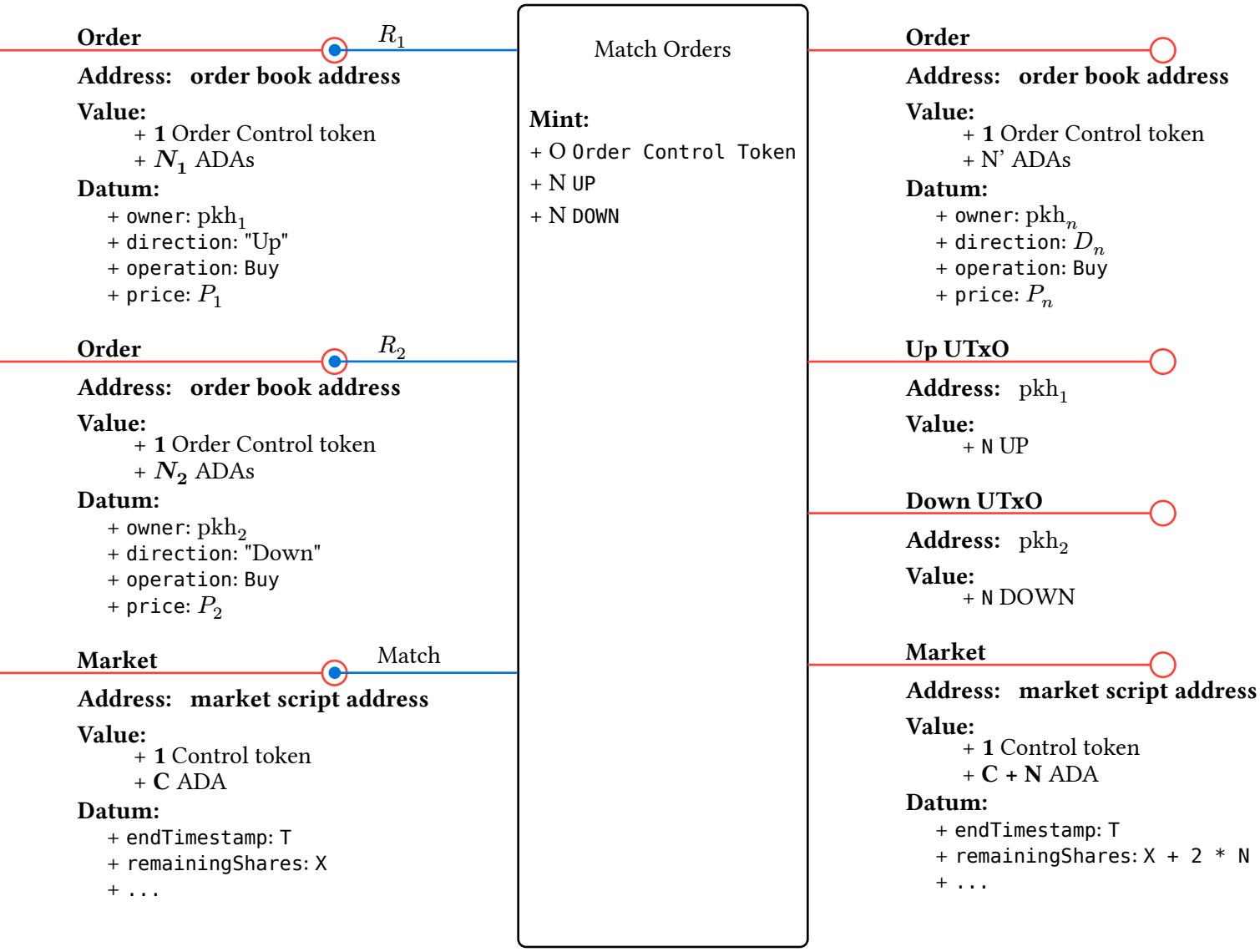

Place Buy Order
Mint:
+ 1 Order Control Token
Order
Address: order book address
Value:
+ 1 Order Control token
+ M ADAs
Datum:
+ owner: pkh
+ direction: "Up|Down"
+ operation: Buy
+ price: N
Place Sell Order
Mint:
+ 1 Order Control Token
Order
Address: order book address
Value:
+ 1 Order Control token
+ M UP/DOWN
Datum:
+ owner: pkh
+ direction: "Up|Down"
+ operation: Sell
+ price: N



Notes: If operation == Sell { M = P * N } else { N = P * M }



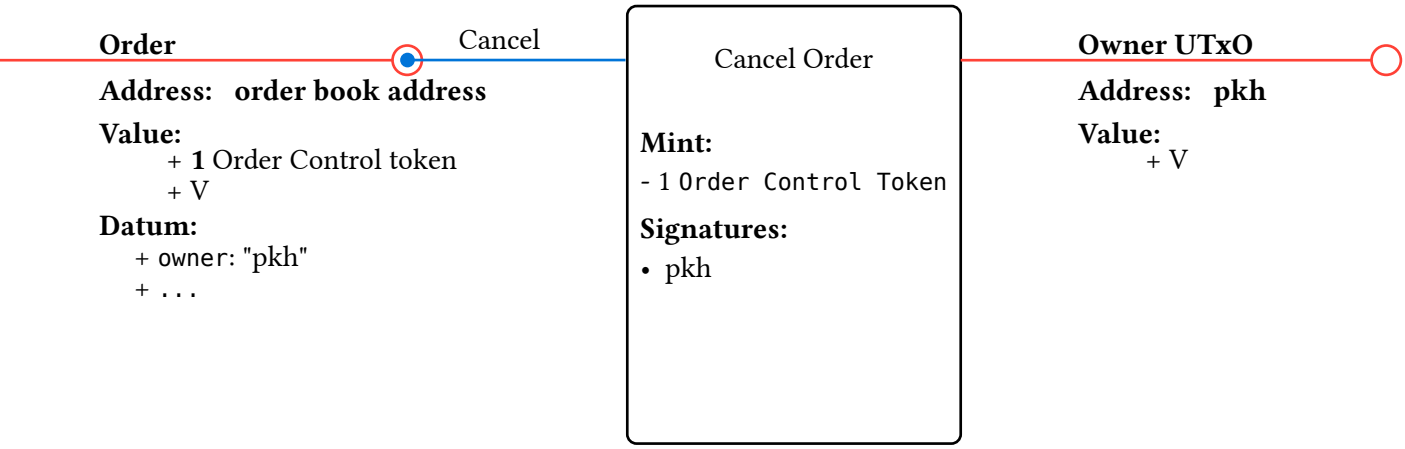
Notes: If operation == Sell { M = P * N' } else { N' = P * M }



Notes:

$$N = \min(N_1, N_2)$$

If N_1 > N_2 { R1 = PartialMatch; R2 = Match; pkh_n = pkh_1; D_n = Up; P_n = P_1; O = −1 }
else if N_2 > N_1 { R1 = Match; R2 = PartialMatch; pkh_n = pkh_2; D_n = Down; P_n = P_2; O = 1}
else { R1 = R2 = Match; O = 2 }


Order
Address: order book address
Value:
+ 1 Order Control token
+ V
Datum:
+ owner: "pkh"
+ ...
Cancel
Cancel Order
Mint:
- 1 Order Control Token
Signatures:
• pkh
Owner UTxO
Address: pkh
Value:
+ V